Sample Size : 6918

वास्तुविस्ता

FUTURE PROSPECTS IN REAL ESTATE PRICE MODELLING

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Introduction

Sample Size : 6918

House price prediction can help the developer determine the selling price of a house with respect to a location and can help the customer to arrange the right time to purchase a house.

There are three factors that influence the price of a house which include physical conditions, concept and location.

- Real estate is least transparent industry in our ecosystem.
- House prices increase every year, so there is a need for a system to predict house prices of the future so that one can gain maximum profitability.
- We aim to make evaluations based on every basic parameter that is considered while determining the price.



Objective

Sample Size : 6918

Develop a House Price Prediction model to provide potential buyers and sellers with the comprehensive insights into the property values. The model will enhance decision-making within the real estate market, enabling more informed transactions and negotiations.

Dataset

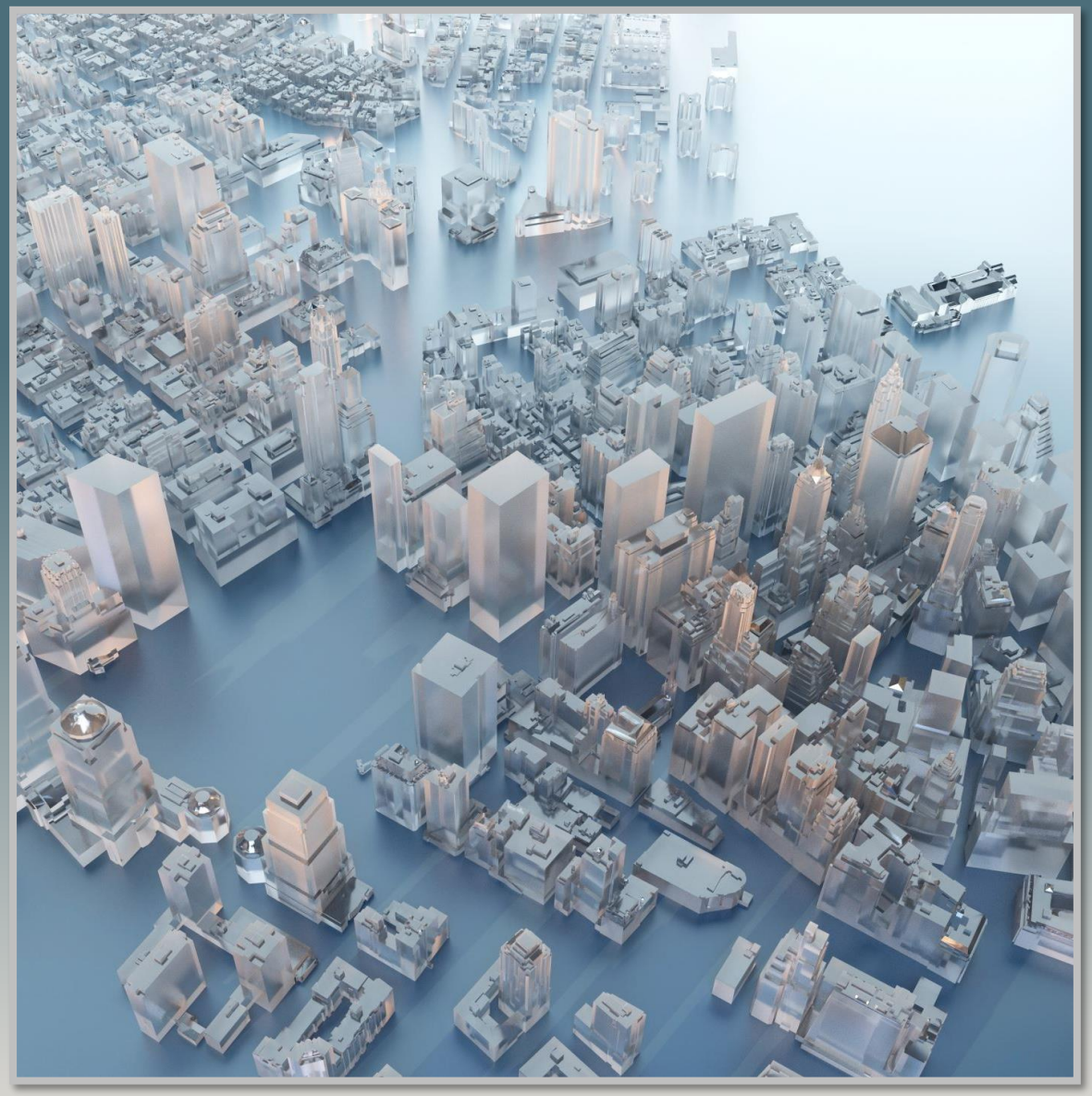
- Source: Kaggle
- Link: https://www.kaggle.com/datasets/amitabhajoy/bengaluru-house-price-data
- Geography: Bengaluru
- No. of Variables: 9
- No. of Observations: 13320
- Tools Used: Python



Variable Description

Sample Size : 6918

Variable Name	Type	Description
Area Type	Categorical	Type of Plot
Availability	Binary	Ready to Move or Not
Location	Categorical	Region of Bangalore
Size	Numeric	BHK
Society	Categorical	Colony in which the House is Present
Total Sq. Ft.	Numeric	Total Area
Bath	Numeric	Number of Bathrooms
Balcony	Numeric	Number of Balconies
Price	Numeric	Cost in Lakhs



Flow Chart

Sample Size : 6918

Exploratory Data Analysis (EDA)

Data Preparation for M.L. Model

- Data Cleaning
- Feature Engineering
- Outlier Removal
- One Hot Encoding

Linear Regression Model Training

Variance Inflation Factor (VIF)

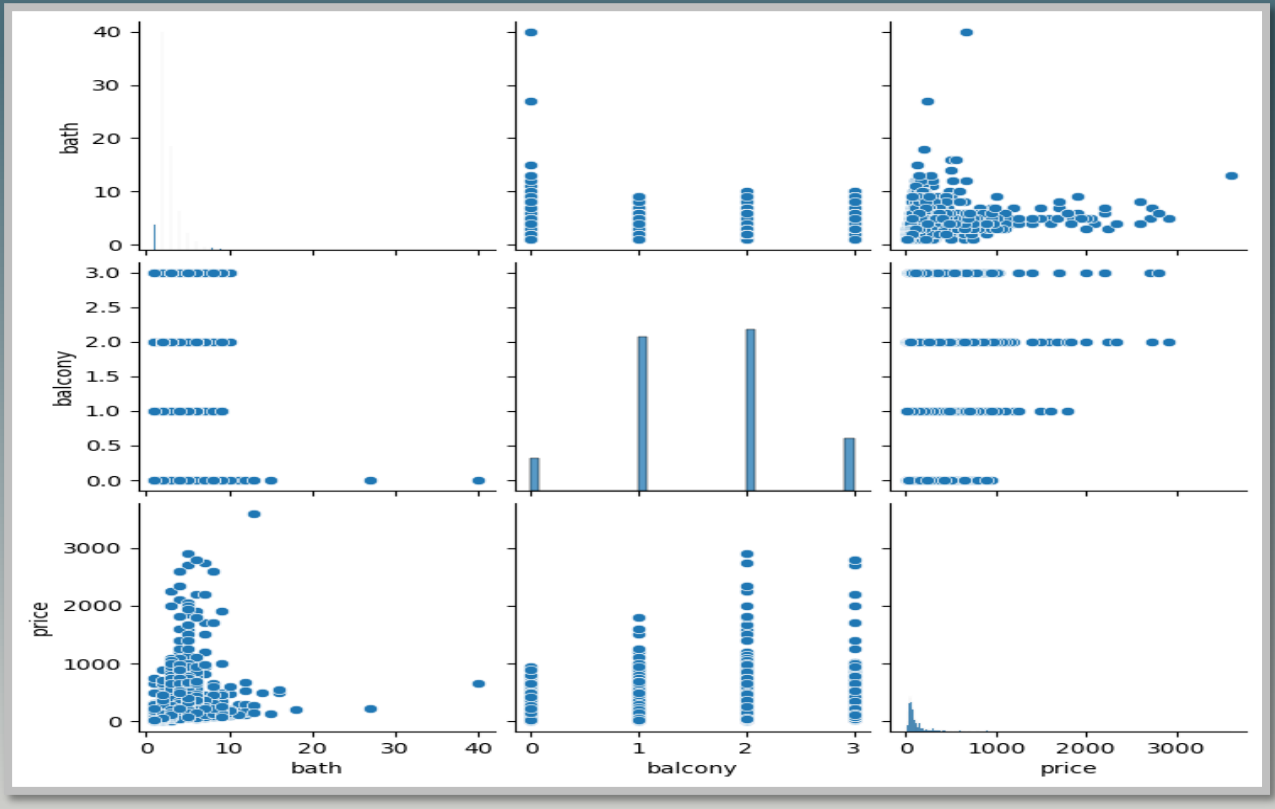
Ridge Regression

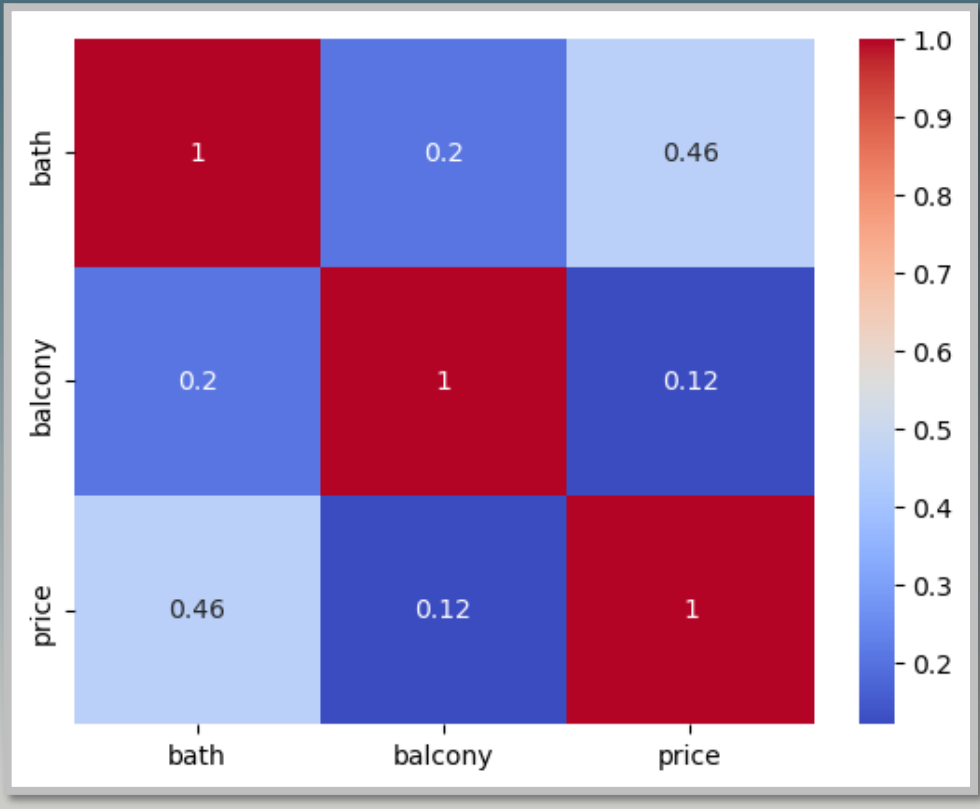
Support Vector Machine (SVM)

Exploratory Data Analysis (EDA)

Sample Size : 6918

Exploratory Data Analysis refers to the critical process of performing initial investigations on data so as to discover patterns, to spot anomalies, to test hypothesis and to check assumptions with the help of summary statistics and graphical representations.





Bath and Price have slightly linear correlation with some outliers

Correlation of Bath is greater than Balcony with price

Data Preparation for ML Model

Sample Size : 6918

Data Cleaning:

Variable	Missing Obs.
Area Type	0
Availability	0
Location	1
Size	16
Society	5502
Total Sq. Ft.	0
Bath	73
Balcony	609
Price	0

As society has 5502 missing observations which is almost 50% of the data. So dropping of Society variable was best possible solution. And for other variables like Location, Size, Bath, Balcony, missing observations were dropped.

Feature Engineering:

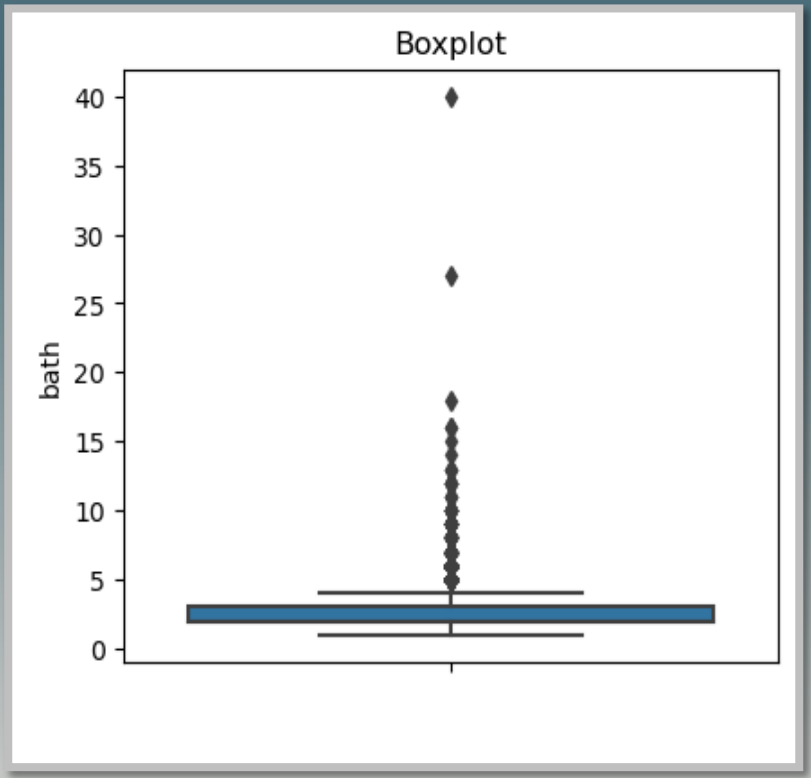
Variable	Obs. Format
Total Sq. Ft.	2830 - 2882
	1752.12
	142.61Sq. Meter
	4125Perch

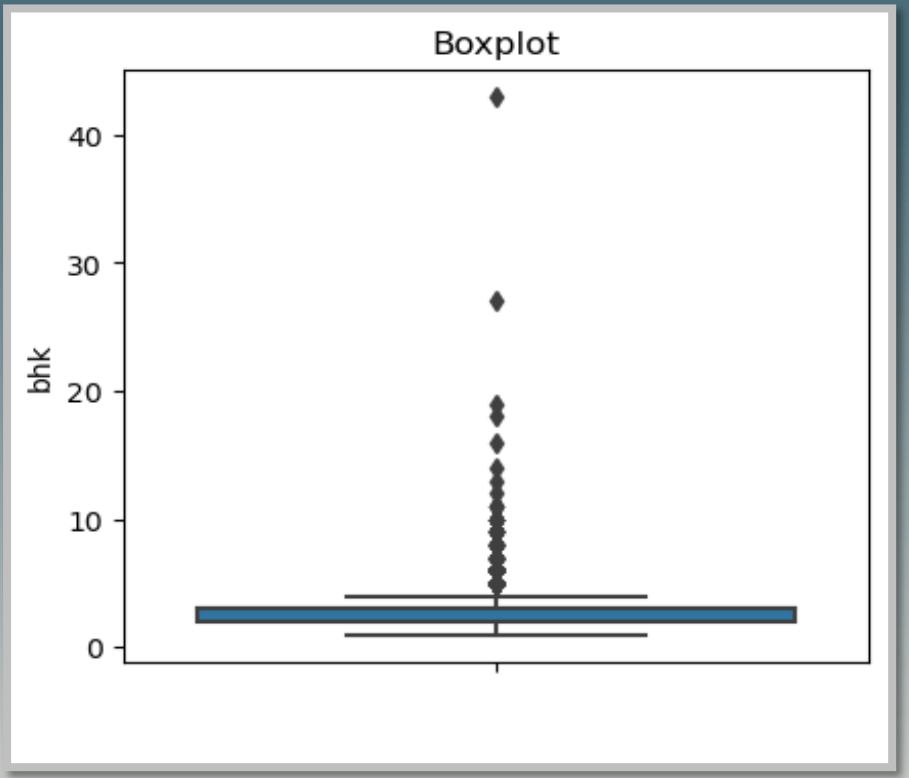
For Total Sq. Ft., the observations which were in range, the mean value was taken and float values were converted to whole number. Whereas observations in string format changed to whole number.

Data Preparation for ML Model

Sample Size : 6918

Outlier Removal:





From the Boxplot, if we observe than there are houses with more than 5 baths. These observations are outlier and can affect the model prediction. Similarly for the bhk variable, houses with more than 5 bhk can be considered as outlier. So those observations were removed from the data.

Data Preparation for ML Model

Sample Size : 6918

One Hot Encoding:

This technique is used to convert the categorical variables into numeric values.

My dataset contains categorical variables which are ‘Area Type’, ‘Location’ and ‘Society’.

After generating the dummy variables for the categorical features, the dataset dimensions transformed from (7120,9) to (7120, 820). Upon applying One-Hot Encoding to these categorical variables, the dataset size shrank to (7120,111). Notably, the ‘Location’ variable was primarily responsible for this substantial dimension shift. Given that the ‘Location’ variable was deemed to have minimal impact on model predictions, it was subsequently excluded from the model.



Linear Regression Model Training

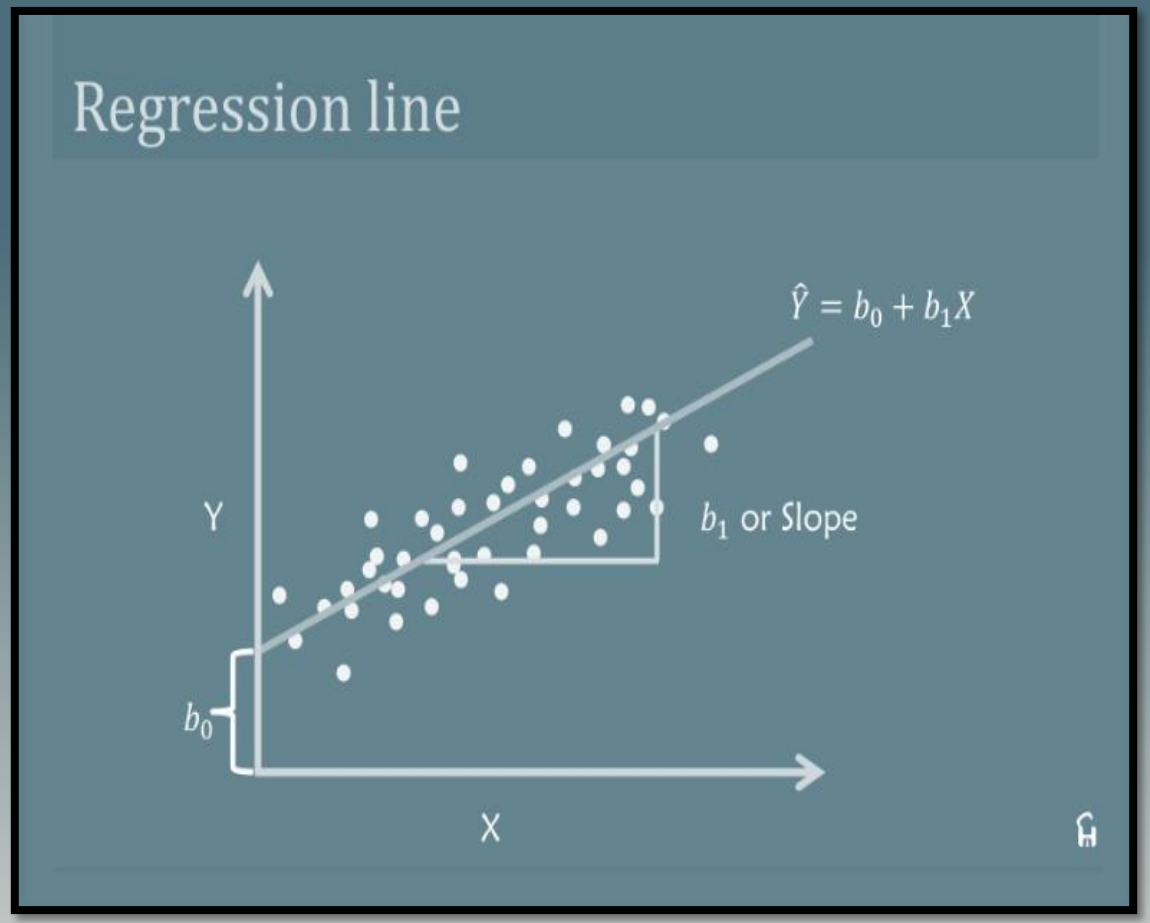
Sample Size : 6918

The process of modelling means training a Machine Learning algorithm to predict the labels from the features.

I have used the Linear Regression algorithm for training and testing of the model.

The R-squared value of 0.8602571050153061 means that The Model is able to explain approximately 86.02% variability in the model.

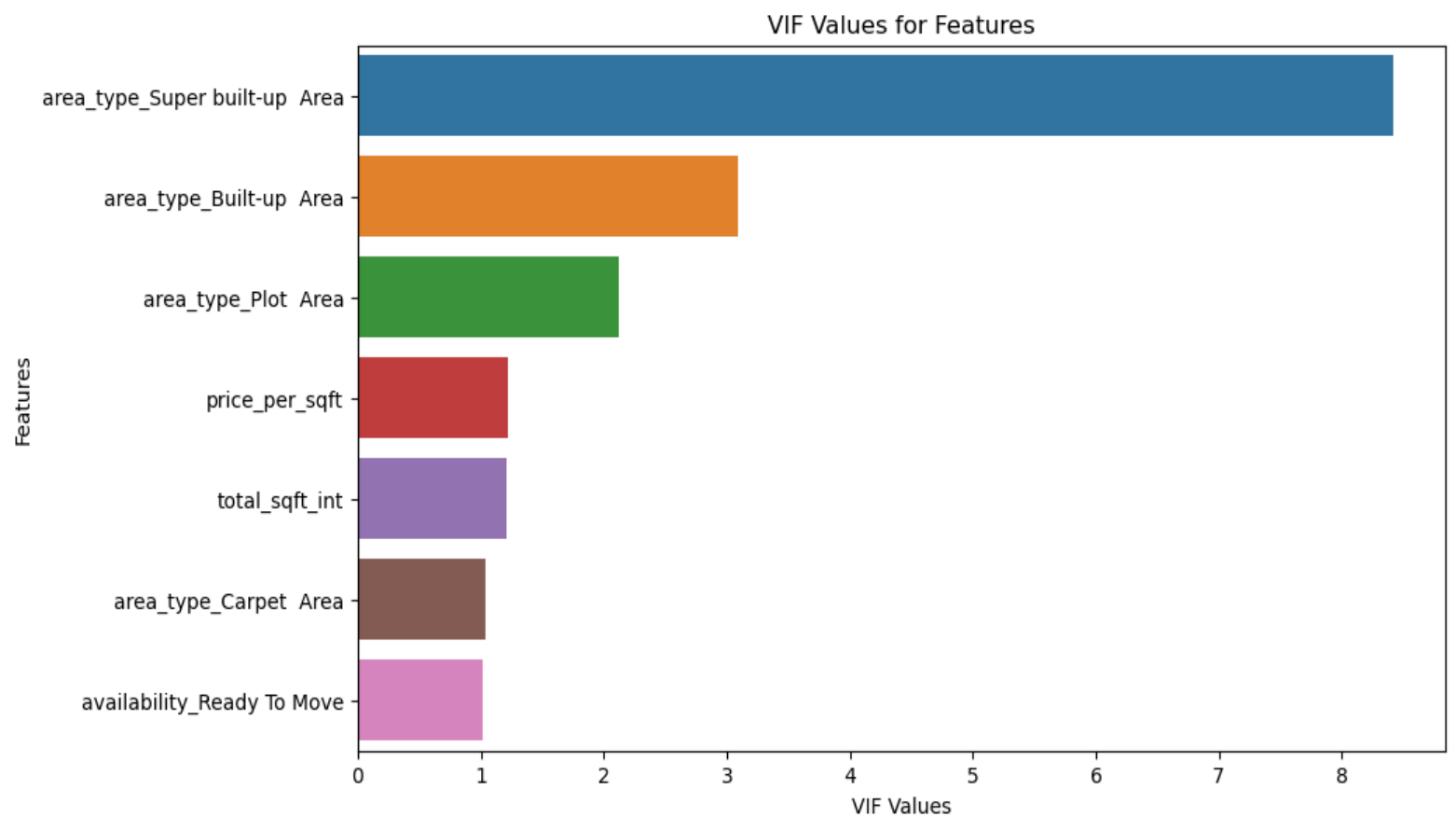
RMSE Score = 37.231422372570734



Model Assumptions

Sample Size : 6918

Variance Inflation Factor



Since VIF value for area_type_Super built-up Area is greater than 5. Which means Multicollinearity is present in the data. So we'll go for Ridge Regression.

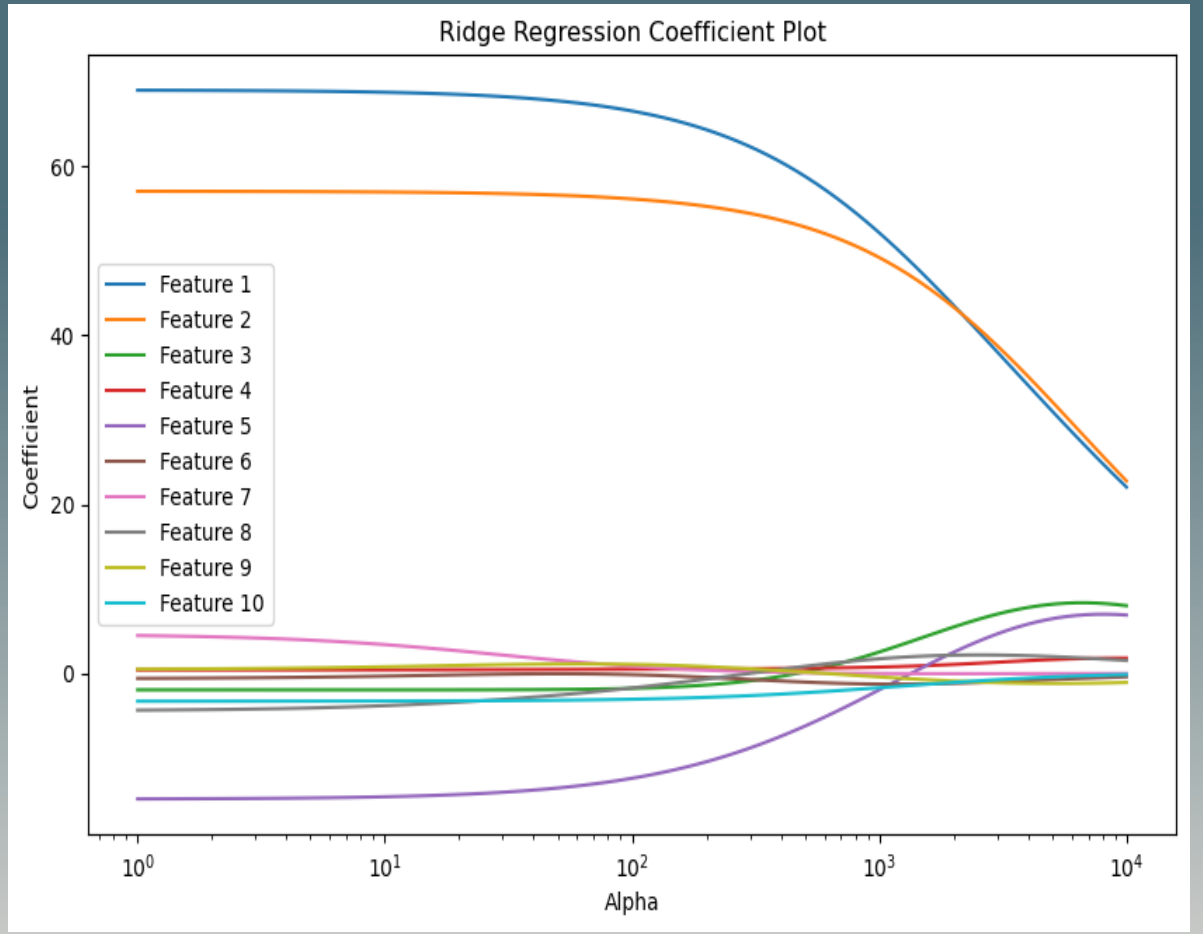
Ridge Regression

Sample Size : 6918

- Ridge Regression is a regularization technique used in Machine Learning model also known as L2 regularization.
- It is used to deal with the problem of Multicollinearity and Overfitting.
- Ridge Regression model adds a penalty term in the cost function of Linear Regression.
- Ridge Regression performs feature selection by shrinking the estimates towards zero.

The R-squared value of 0.8602756801117902 means that The model is able to explain approximately 86.02% variability in the model. Which is same as Linear Regression model prediction.

RMSE Score = 37.228947827277715



Model Coefficients

Sample Size : 6918

Variable	LR Coefficients	Ridge Coefficients
β_0	135.21934894971616	135.1027543939328
bath	69.02320245	68.99651273
balcony	57.04929496	57.03968734
total_sqft_int	-1.95670271	-1.9573933
Bhk	0.39827991	0.39980158
price_per_sqft':	-14.89272858	-14.86397877
area_type_Built-up	-0.64326498	-0.60084243
area_type_Carpet Area	4.63046935	4.47106526
area_type_Plot Area	-4.43651262	-4.36283194
area_type_Super built-up Area	0.44930825	0.4926091
availability_Ready To Move	-3.30283139	-3.30081223

Ridge Regression

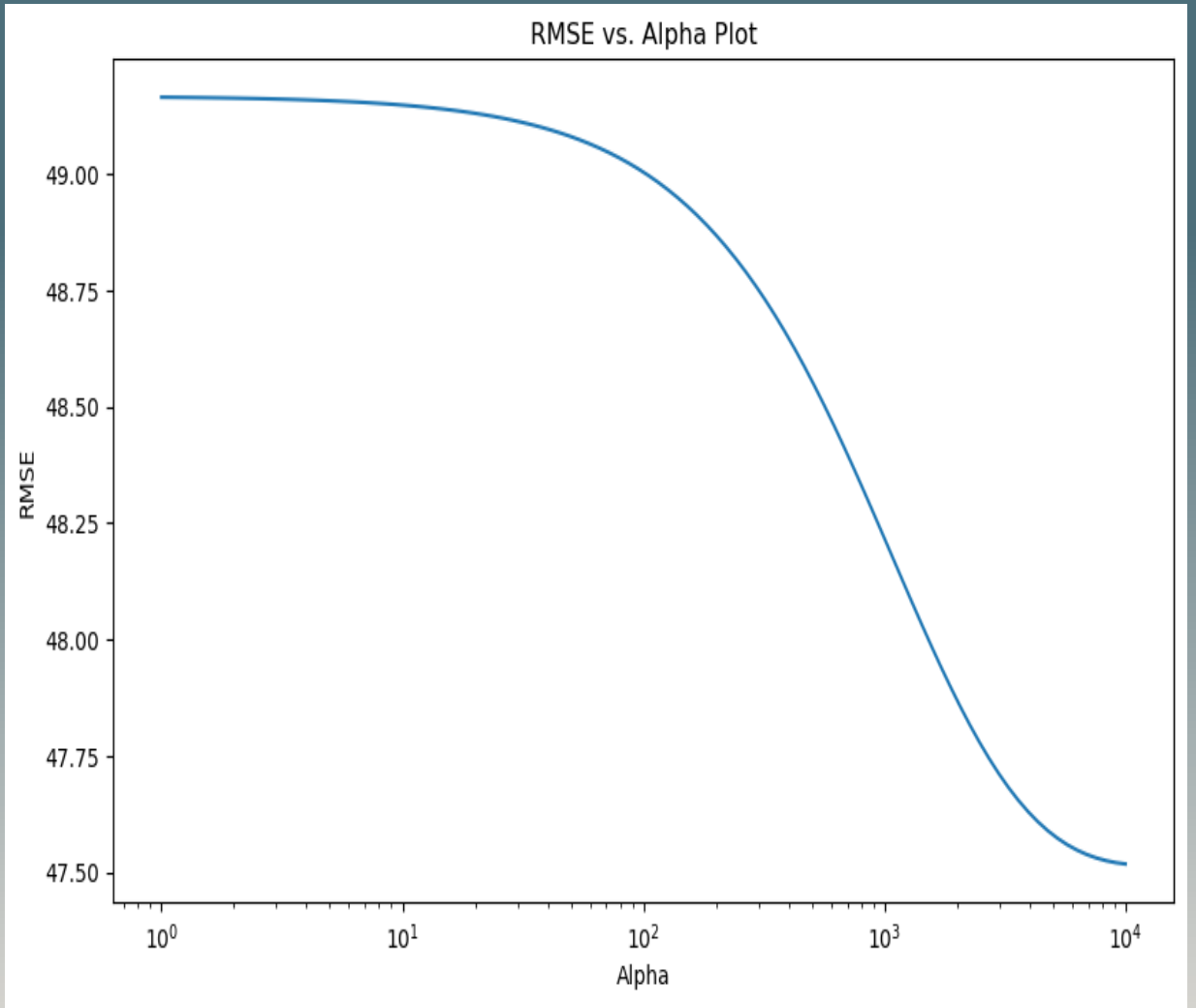
Sample Size : 6918

Hyperparameter Tunning

After performing the Hyperparameter selection, I got the Best Alpha as 10000 .

When run the model with best alpha. The model performance score was 0.859500457127703 which is 86.0%.

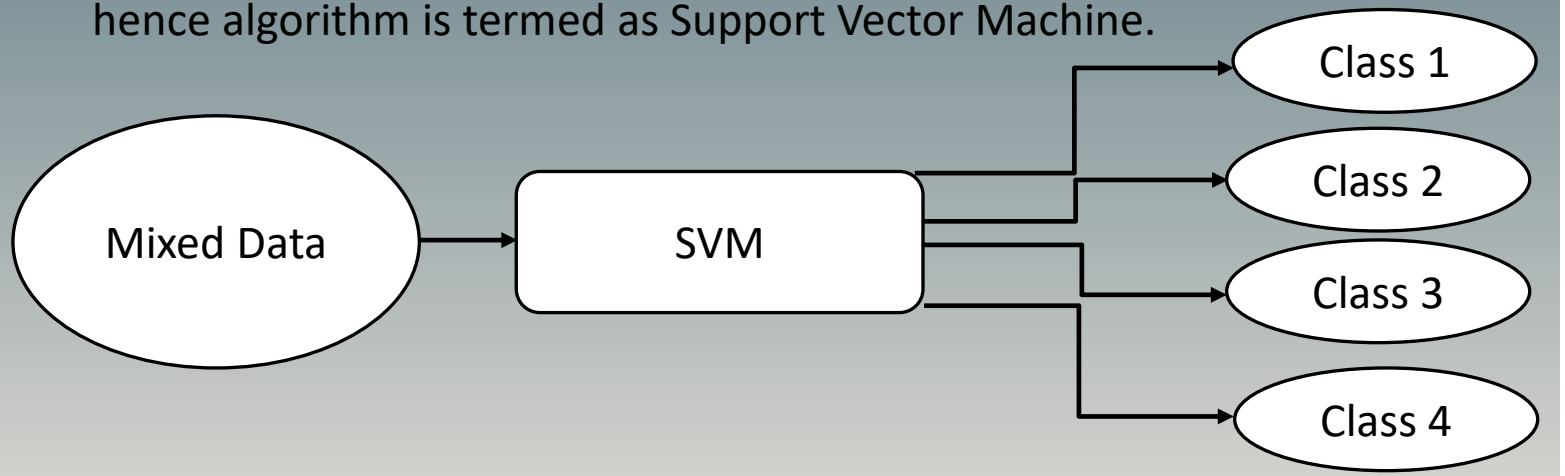
RMSE Score = 37.33208239684906

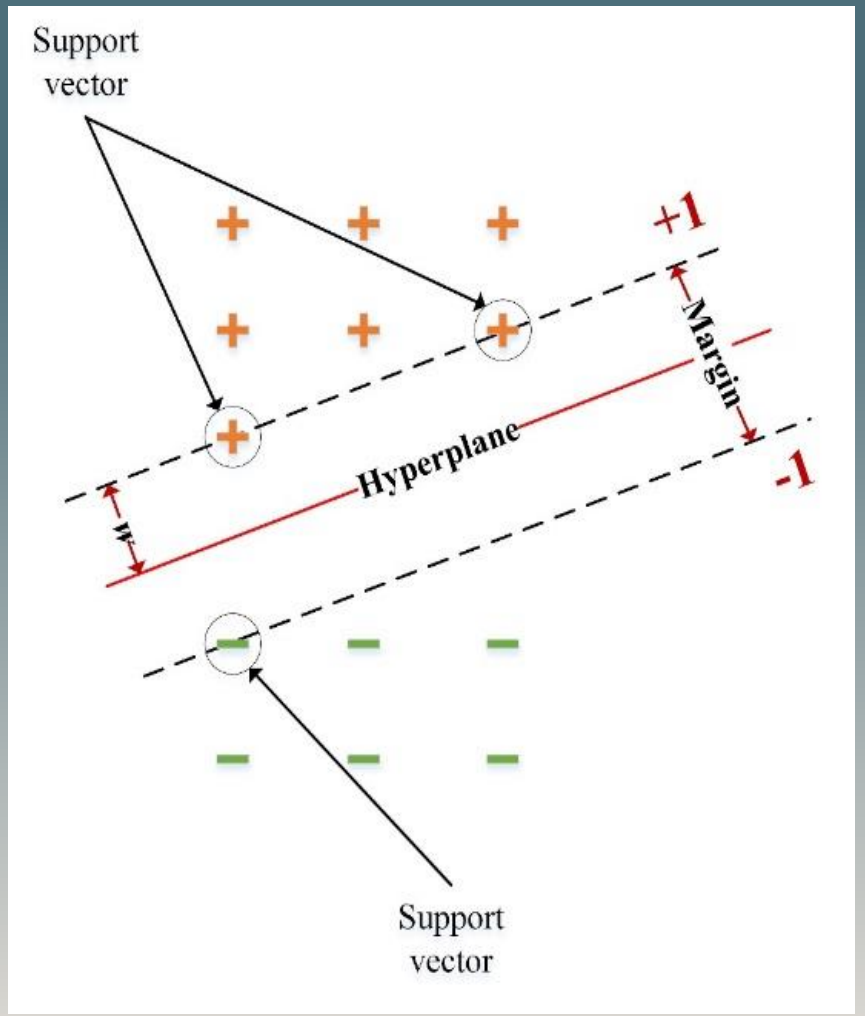


Support Vector Machine (SVM)

Sample Size : 6918

- Support Vector Machine or SVM is a Supervised Learning algorithms, which is used for Classification as well as Regression problems in Machine Learning.
- The goal of the SVM is to create a decision boundary that can segregate n-dimensional space into classes so that we can easily put the new data point in the correct category in future. The best decision boundary is called a Hyperplane.
- SVM chooses the extreme points/vectors that helps in creating the Hyperplane. These extreme points are called as Support Vectors, and hence algorithm is termed as Support Vector Machine.





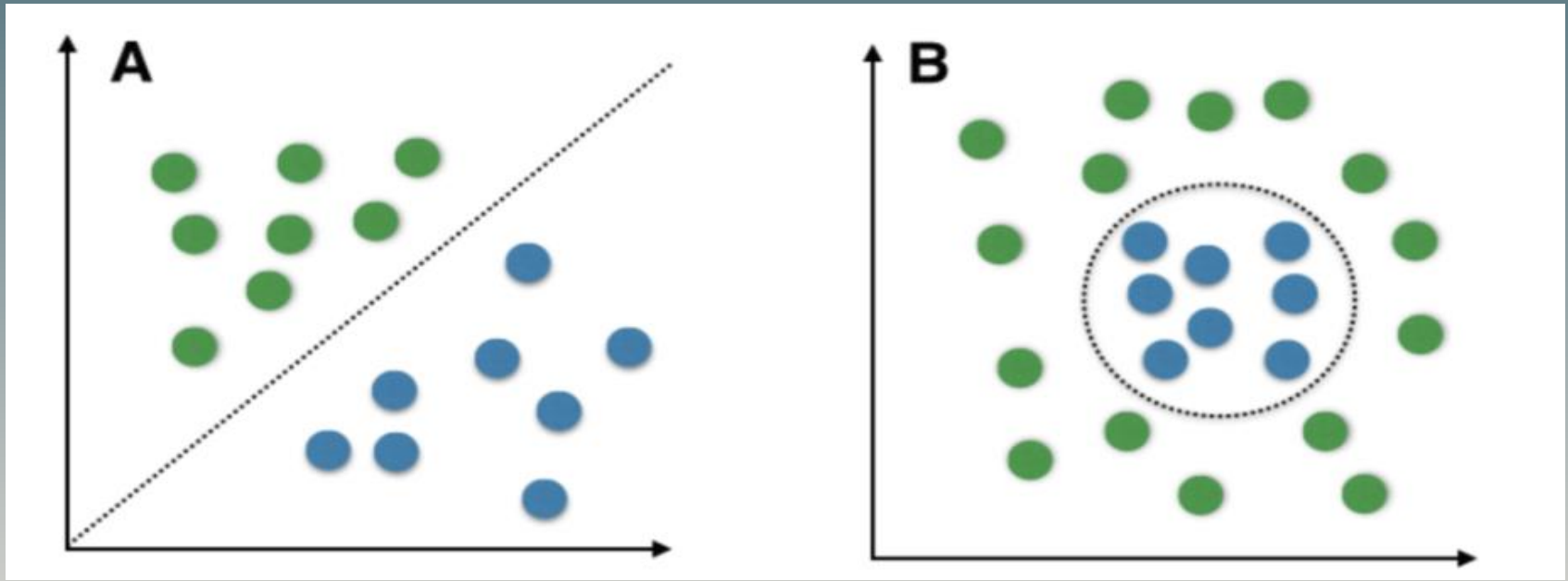
SVM Classifiers

Sample Size : 6918

There are 2 kinds of SVM Classifiers

1. Linear SVM Classifier
2. Non-Linear SVM Classifier

Linear vs Non-Linear

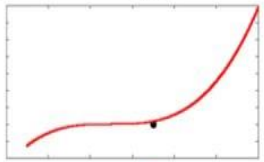


Tunning Parameters: Kernal

Sample Size : 6918

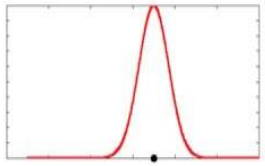
Common kernel functions

- Some commonly used kernel functions & their shape:
- Polynomial $K(a,b) = (1 + \sum_j a_j b_j)^d$



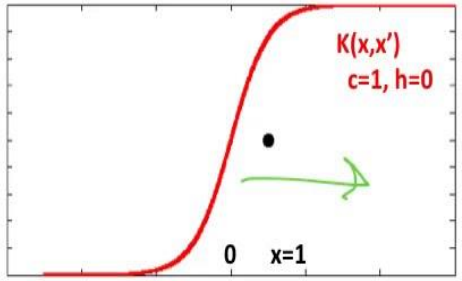
- Radial Basis Functions

$$K(a,b) = \exp(-(a-b)^2/2\sigma^2)$$

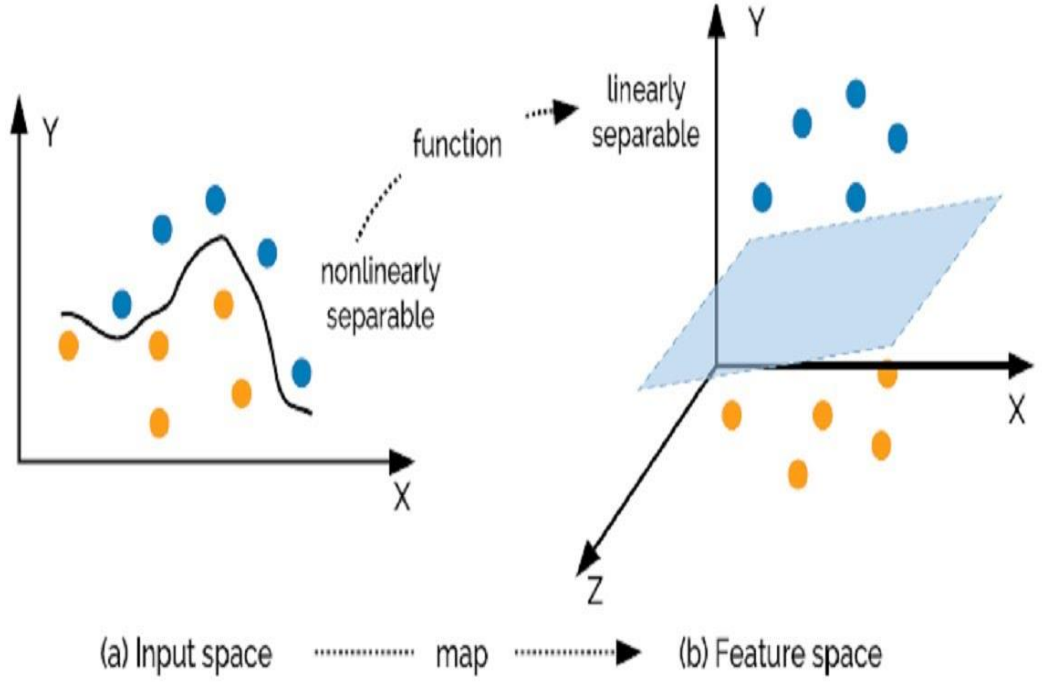


- Saturating, sigmoid-like:

$$K(a,b) = \tanh(ca^T b + h)$$



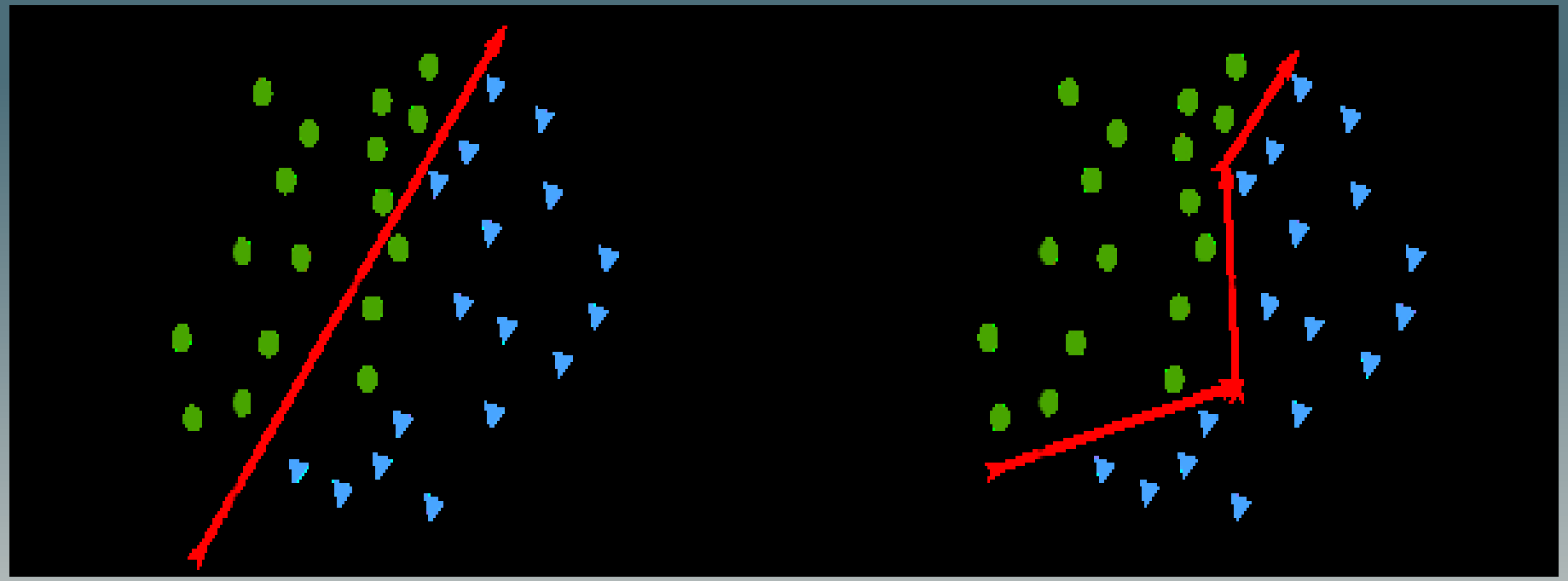
Kernal Trick (SVM)...



Tunning Parameters: Regularization

Sample Size : 6918

To avoid misclassification of each training example



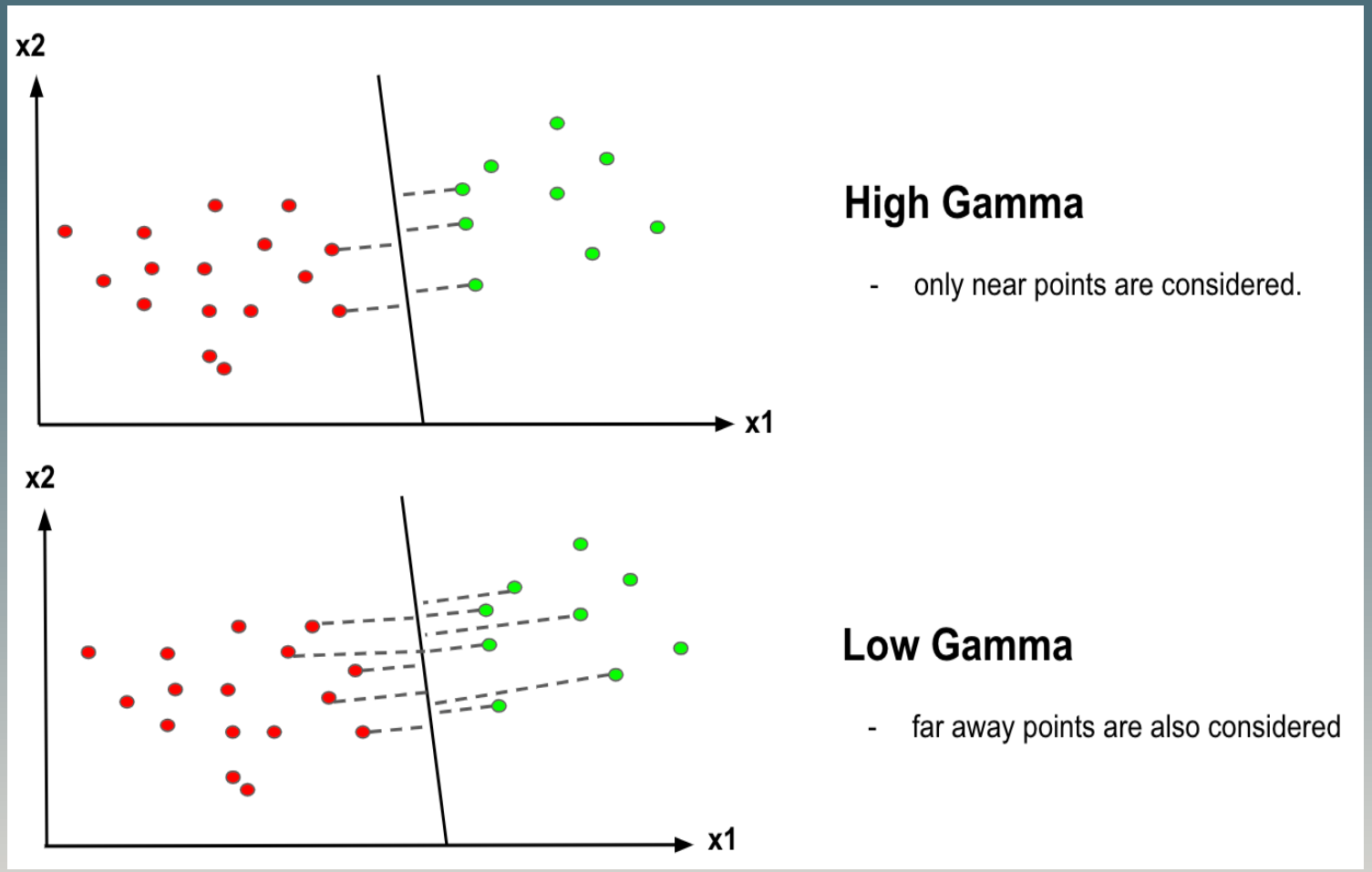
Low Regularization Value

High Regularization Value

Tunning Parameters: Gamma

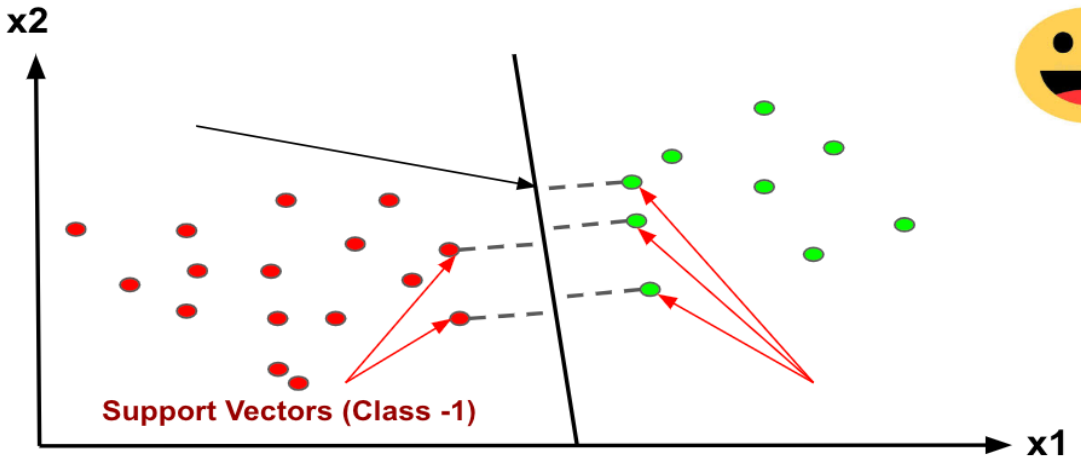
Sample Size : 6918

Gamma is a hyperparameter that controls the flexibility and complexity of the model's decision boundary. Gamma determines the range of influence of individual data points on the shape of the decision boundary.



Tunning Parameters: Margin

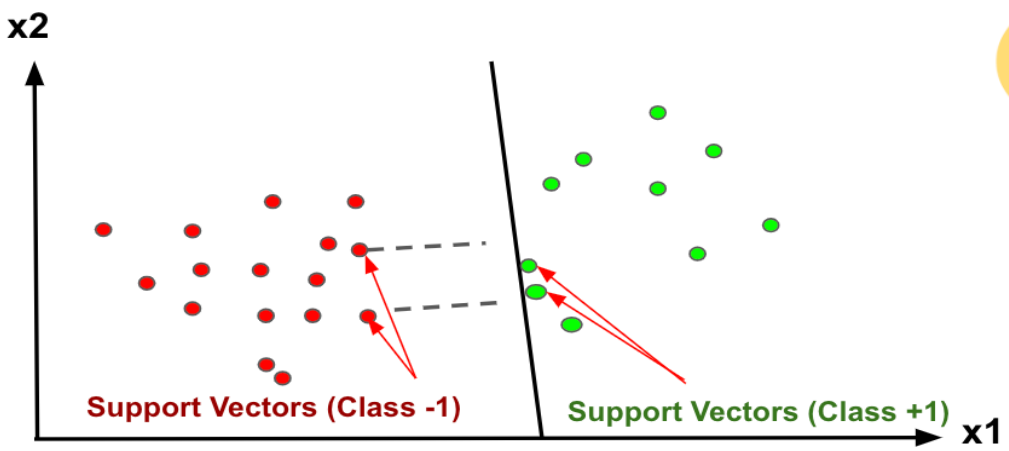
Sample Size : 6918





Good Margin

- all sector vectors have the same distance with the maximum margin hyperplane





Bad Margin

- very close to either class -1 support vectors or class +1 support vectors

Performance Metrics

Sample Size : 6918

Kernal	Best Hyper Parameters	RMSE	R-Square
linear	{'C': 100, 'gamma': 0.001}	43.26665243399351	0.8112803729644541
rbf	{'C': 100, 'gamma': 0.01}	35.68954014031426	0.8715919106183342
sigmoid	{'C': 100, 'gamma': 0.001}	46.74849372142302	0.7796841593292427

Looking at the RMSE and R-Square values of the SVM model with different kernel. We can conclude that the SVM model with ‘rbf’ (Radial Basis Function) kernel is performing better on this data as the RMSE = 35.7 indicating it has the smallest average prediction error and R-sq = 87.16% suggesting that the model is able to explain the variance relatively well in my data.

Thank You!